Calibration and Hallucination Rates in 7B to 9B Open-Source LLMs.

Ji et al. (2023) categorized natural language generation failures into intrinsic hallucinations that contradict source text and extrinsic hallucinations that invent unverifiable claims. Kadavath et al. (2022) showed that smaller parameter language models are poorly calibrated at self-evaluation and exhibit high surface confidence when stating falsehoods.

Bang et al. (2025) evaluated 7B to 8B open-weight models including Llama-3.1-8B, Mistral-7B, and Qwen-2.5-7B on the HalluLens benchmark, recording baseline hallucination rates between 48% and 85%. Chuang et al. (2024) explained via Lookback Lens that attention gradually shifts away from source context toward newly generated tokens during long-form summarization.